

Orthonormal Basis (Tetrad) Methods for Computing Curvature

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July 29, 2026

Summary

This note provides a self-contained derivation of the Riemann curvature tensor and the Ricci tensor in an orthonormal basis, using the method of Ricci rotation coefficients. The basic framework of the orthonormal tetrad formalism is first reviewed, followed by detailed proofs of the equivalence between metric compatibility and the antisymmetry of the connection one-forms, and between torsion-freeness and the commutation relations of the basis vectors. Explicit component formulas for the Riemann tensor in terms of the Ricci rotation coefficients are derived, and their translation into differential forms notation is also demonstrated.

Keywords: Orthonormal basis, Ricci rotation coefficients, Riemann curvature tensor, differential forms, tetrad formalism

1 Orthonormal Basis

We introduce a set of smooth vector fields $(e_\mu)^a$ as a noncoordinate, orthonormal basis satisfying

$$(e_\mu)^a (e_\nu)_a = \eta_{\mu\nu}, \quad (1)$$

where $\eta_{\mu\nu} = \text{diag}(-1, \dots, -1, 1, \dots, 1)$.

Theorem 1.1. *Equation (1) implies the useful relation*

$$\eta^{\mu\nu} (e_\mu)^a (e_\nu)_b = \delta^a_b, \quad (2)$$

where δ^a_b is the identity map on vectors and $\eta^{\mu\nu}$ is the inverse of $\eta_{\mu\nu}$ (so that $\eta^{\mu\nu} = \eta_{\mu\nu}$).

Proof. Contracting the left-hand side of the above equation with an arbitrary basis vector $(e_\sigma)^b$, we obtain

$$\eta^{\mu\nu} (e_\mu)^a (e_\nu)_b (e_\sigma)^b = \eta^{\mu\nu} (e_\mu)^a \eta_{\nu\sigma} = (e_\mu)^a \delta^\mu_\sigma = (e_\sigma)^a. \quad (3)$$

The right-hand side gives $\delta^a_b (e_\sigma)^b = (e_\sigma)^a$, which is equal to the left-hand side. Thus the theorem is proved. $\square$

2 Consequences of $\nabla_a g_{bc} = 0$

We begin by defining the connection *l-forms*, $\omega_{a\mu\nu}$, by

$$\omega_{a\mu\nu} = (e_\mu)^b \nabla_a (e_\nu)_b \quad . \quad (4)$$

The components

$$\omega_{\lambda\mu\nu} = (e_\lambda)^a (e_\mu)^b \nabla_a (e_\nu)_b \quad . \quad (5)$$

of $\omega_{a\mu\nu}$ are called the *Ricci rotation coefficients*.

Theorem 2.1. *Metric compatibility of the derivative operator is equivalent to $\omega_{a\mu\nu} = -\omega_{a\nu\mu}$.*

Proof. From $\nabla_a g_{bc} = 0$, we have

$$\nabla_a [g_{bc} (e_\mu)^c (e_\nu)^b] = (e_\mu)^c (e_\nu)^b \nabla_a g_{bc} + g_{bc} \nabla_a [(e_\mu)^c (e_\nu)^b] = g_{bc} \nabla_a [(e_\mu)^c (e_\nu)^b] \quad (6)$$

But

$$\nabla_a [g_{bc} (e_\mu)^c (e_\nu)^b] = \nabla_a (e_\mu)_b (e_\nu)^b = \nabla_a \eta_{\mu\nu} = 0. \quad (7)$$

Thus

$$g_{bc} \nabla_a [(e_\mu)^c (e_\nu)^b] = g_{bc} (e_\mu)^c \nabla_a (e_\nu)^b + g_{bc} (e_\nu)^b \nabla_a (e_\mu)^c = 0. \quad (8)$$

Again using $\nabla_a g_{bc} = 0$ together with the definition (4), the above equation is equivalent to

$$\omega_{a\mu\nu} = (e_\mu)^b \nabla_a (e_\nu)_b = -(e_\nu)^b \nabla_a (e_\mu)_b = -\omega_{a\nu\mu}. \quad (9)$$

Conversely, from $\omega_{a\mu\nu} = -\omega_{a\nu\mu}$, one can also derive $\nabla_a g_{bc} = 0$. Therefore

$$\nabla_a g_{bc} = 0 \quad \Leftrightarrow \quad \omega_{a\mu\nu} = -\omega_{a\nu\mu}. \quad (10)$$

This completes the proof. $\square$

3 Consequences of Torsion-Freeness

Theorem 3.1. *Torsion-freeness is equivalent to*

$$(1) (e_\sigma)_a [e_\mu, e_\nu]^a = \omega_{\mu\sigma\nu} - \omega_{\nu\sigma\mu} \quad \text{or} \quad (2) \partial_{[a} (e_\sigma)_{b]} = \eta^{\mu\nu} (e_\mu)_{[a} \omega_{b]\sigma\nu}. \quad (11)$$

Proof. (1) We first need to prove the following lemma: For arbitrary vector fields v^a and w^a , the commutator formula

$$[v, w]^b = v^a \nabla_a w^b - w^a \nabla_a v^b \quad (12)$$

is equivalent to torsion-freeness. Suppose first that ∇_a is torsion-free, and let f be an arbitrary smooth scalar field. Then

$$\begin{aligned} [u, v](f) &= u(v(f)) - v(u(f)) \\ &= u^b \nabla_b (v^a \nabla_a f) - v^b \nabla_b (u^a \nabla_a f) \\ &= u^b (\nabla_b v^a) \nabla_a f + v^a u^b \nabla_b \nabla_a f - v^b (\nabla_b u^a) \nabla_a f - u^a v^b \nabla_b \nabla_a f \\ &= (u^b \nabla_b v^a - v^b \nabla_b u^a) \nabla_a f, \end{aligned} \quad (13)$$

where in the fourth step we used the torsion-free condition. Also, since

$$[u, v](f) = [u, v]^a \nabla_a f \quad (14)$$

and f is arbitrary, torsion-freeness implies

$$[v, w]^b = v^a \nabla_a w^b - w^a \nabla_a v^b. \quad (15)$$

Conversely, the above formula readily implies torsion-freeness, so they are equivalent. Using this lemma, it suffices to prove that

$$[e_\mu, e_\nu]^a = (e_\mu)^b \nabla_b (e_\nu)^a - (e_\nu)^b \nabla_b (e_\mu)^a \quad (16)$$

is equivalent to

$$(e_\sigma)_a [e_\mu, e_\nu]^a = \omega_{\mu\sigma\nu} - \omega_{\nu\sigma\mu}. \quad (17)$$

This is immediate, since contracting the first equation with $(e_\sigma)_a$ and using Eq. (4) gives

$$(e_\sigma)_a [e_\mu, e_\nu]^a = (e_\sigma)_a \{ (e_\mu)^b \nabla_b (e_\nu)^a - (e_\nu)^b \nabla_b (e_\mu)^a \} = \omega_{\mu\sigma\nu} - \omega_{\nu\sigma\mu}. \quad (18)$$

Conversely, the first formula follows easily from the second and Eq. (4).

(2) Before proving the second equivalent form, we also need to prove the following lemma: torsion-freeness is equivalent to $C^c_{ab} = C^c_{ba}$. Let $\omega_b = \nabla_b f = \tilde{\nabla}_b f$, where $f \in \mathcal{F}_M$. Then the formula

$$\nabla_a T^{b_1 \dots b_k}_{c_1 \dots c_l} = \tilde{\nabla}_a T^{b_1 \dots b_k}_{c_1 \dots c_l} + \sum_i C^{b_i}_{ad} T^{b_1 \dots d \dots b_k}_{c_1 \dots c_l} - \sum_j C^d_{ac_j} T^{b_1 \dots b_k}_{c_1 \dots d \dots c_l} \quad (19)$$

gives

$$\nabla_a \nabla_b f = \tilde{\nabla}_a \tilde{\nabla}_b f - C^c_{ab} \nabla_c f. \quad (20)$$

Interchanging a and b , we obtain

$$\nabla_b \nabla_a f = \tilde{\nabla}_b \tilde{\nabla}_a f - C^c_{ba} \nabla_c f. \quad (21)$$

Subtracting the two equations and using the torsion-free condition, we have

$$C^c_{ab} \nabla_c f = C^c_{ba} \nabla_c f. \quad (22)$$

Let $T^c_{ab} \equiv C^c_{ab} - C^c_{ba}$. Then for all $f \in \mathcal{F}_M$, we have $T^c_{ab} \nabla_c f = 0$. Hence, in any coordinate basis, the components of T^c_{ab} are

$$T^\sigma_{\mu\nu} = T^c_{ab} (dx^\sigma)_c (\partial/\partial x^\mu)^a (\partial/\partial x^\nu)^b = 0, \quad (23)$$

where the second equality follows from $T^c_{ab} (dx^\sigma)_c = T^c_{ab} \nabla_c x^\sigma = 0$ (taking x^σ as f). Therefore $T^c_{ab} = 0$. Conversely, the torsion-free condition also implies $C^c_{ab} = C^c_{ba}$, so they are equivalent. Using this lemma, it suffices to prove that $C^c_{ab} = C^c_{ba}$ is equivalent to

$$\partial_{[a} (e_{\sigma]b}) = \eta^{\mu\nu} (e_\mu)_{[a} \omega_{b]\sigma\nu}. \quad (24)$$

This is straightforward, since

$$\nabla_{[a}(e_{\sigma})_{b]} = \eta^{\mu\nu}(e_{\mu})_{[a}\omega_{b]\sigma\nu}, \quad (25)$$

and

$$\nabla_{[a}(e_{\sigma})_{b]} = \partial_{[a}(e_{\sigma})_{b]} + \Gamma^c_{[ab]}(e_{\sigma})_c. \quad (26)$$

This completes the proof. $\square$

Corollary 3.2. *From the antisymmetry property $\omega_{a\mu\nu} = -\omega_{a\nu\mu}$ established in Eq. (9), we have the decomposition*

$$\omega_{\lambda\mu\nu} = 3\omega_{[\lambda\mu\nu]} - 2\omega_{[\mu\nu]\lambda}. \quad (27)$$

Substituting this into Eq. (18), we obtain an expression for $\omega_{\lambda\mu\nu}$ solely in terms of the commutators (or antisymmetrized derivatives) of the orthonormal basis vectors:

$$\omega_{\lambda\mu\nu} = \frac{1}{2} \left[(e_{\lambda})_a [e_{\mu}, e_{\nu}]^a + (e_{\mu})_a [e_{\nu}, e_{\lambda}]^a - (e_{\nu})_a [e_{\lambda}, e_{\mu}]^a \right]. \quad (28)$$

This yields the $n^2(n-1)/2$ independent equations needed to solve for the Ricci rotation coefficients.

Proof. By Theorem 2.1, $\omega_{a\mu\nu}$ is antisymmetric in its last two indices:

$$\omega_{a\mu\nu} = -\omega_{a\nu\mu}. \quad (29)$$

For any such tensor, the following algebraic identity holds:

$$\omega_{\lambda\mu\nu} = 3\omega_{[\lambda\mu\nu]} - 2\omega_{[\mu\nu]\lambda}. \quad (30)$$

Because

$$\begin{aligned} 3\omega_{[\lambda\mu\nu]} - 2\omega_{[\mu\nu]\lambda} &= \frac{1}{2} (\omega_{\lambda\mu\nu} - \omega_{\mu\lambda\nu} + \omega_{\mu\nu\lambda} - \omega_{\nu\mu\lambda} + \omega_{\nu\lambda\mu} - \omega_{\lambda\nu\mu}) - \frac{1}{2} (\omega_{\mu\nu\lambda} - \omega_{\nu\mu\lambda}) \\ &= \omega_{\lambda\mu\nu}, \end{aligned} \quad (31)$$

where the last step uses (29).

From Eq. (18), we have

$$(e_{\sigma})_a [e_{\mu}, e_{\nu}]^a = \omega_{\mu\sigma\nu} - \omega_{\nu\sigma\mu}. \quad (32)$$

Using (29), the right-hand side becomes

$$\omega_{\mu\sigma\nu} - \omega_{\nu\sigma\mu} = -2\omega_{[\mu\nu]\sigma}. \quad (33)$$

Substituting (33) into (32) gives

$$(e_{\sigma})_a [e_{\mu}, e_{\nu}]^a = -2\omega_{[\mu\nu]\sigma}. \quad (34)$$

Equation (30) gives

$$\omega_{\lambda\mu\nu} = 3\omega_{[\lambda\mu\nu]} - 2\omega_{[\mu\nu]\lambda}$$

$$\begin{aligned}
&= \omega_{[\mu\nu]\lambda} + \omega_{[\nu\lambda]\mu} + \omega_{[\lambda\mu]\nu} - 2\omega_{[\mu\nu]\lambda} \\
&= \omega_{[\nu\lambda]\mu} + \omega_{[\lambda\mu]\nu} - \omega_{[\mu\nu]\lambda}.
\end{aligned} \tag{35}$$

Comparing this with Eq. (34), we see that

$$\omega_{\lambda\mu\nu} = \frac{1}{2} \left[(e_\lambda)_a [e_\mu, e_\nu]^a + (e_\mu)_a [e_\nu, e_\lambda]^a - (e_\nu)_a [e_\lambda, e_\mu]^a \right]. \tag{36}$$

$\omega_{\lambda\mu\nu}$ has three indices, but due to the antisymmetry $\omega_{\lambda\mu\nu} = -\omega_{\lambda\nu\mu}$, the last two indices cannot be equal in nonzero components. Thus there are n choices for the first index, and the number of choices for the last two indices is

$$\binom{n}{2} = \frac{n(n-1)}{2}. \tag{37}$$

Hence the number of independent components is

$$n \cdot \frac{n(n-1)}{2} = \frac{n^2(n-1)}{2}. \tag{38}$$

Each commutator $(e_\lambda)_a [e_\mu, e_\nu]^a$ on the right-hand side of Eq. (36) is also antisymmetric in μ and ν , so the right-hand side also has

$$n \cdot \frac{n(n-1)}{2} = \frac{n^2(n-1)}{2} \tag{39}$$

independent components. Therefore, the left-hand side has $\frac{n^2(n-1)}{2}$ independent unknowns, and the right-hand side provides the same number of independent equations. This yields the $n^2(n-1)/2$ independent equations needed to solve for the Ricci rotation coefficients. $\square$

4 Expression for the Riemann Curvature Tensor

4.1 Tensor Representation

The Riemann tensor can be expressed in terms of the Ricci rotation coefficients in the following manner. Since

$$R_{abc}{}^d \omega_d = \nabla_a \nabla_b \omega_c - \nabla_b \nabla_a \omega_c, \tag{40}$$

we have

$$R_{abcd} \omega^d = \nabla_a \nabla_b \omega_c - \nabla_b \nabla_a \omega_c. \tag{41}$$

Taking ω^d to be the basis vector $(e_\nu)^d$, we obtain the components of R_{abcd} in our orthonormal basis:

$$\begin{aligned}
R_{\rho\sigma\mu\nu} &= R_{abcd} (e_\rho)^a (e_\sigma)^b (e_\mu)^c (e_\nu)^d \\
&= (e_\rho)^a (e_\sigma)^b (e_\mu)^c (\nabla_a \nabla_b - \nabla_b \nabla_a) (e_\nu)_c.
\end{aligned} \tag{42}$$

Now we compute the right-hand side. First, using the product rule,

$$(e_\mu)^c \nabla_a \nabla_b (e_\nu)_c = \nabla_a \left[(e_\mu)^c \nabla_b (e_\nu)_c \right] - \left[\nabla_a (e_\mu)^c \right] \left[\nabla_b (e_\nu)_c \right]$$

$$= \nabla_a \omega_{b\mu\nu} - [\nabla_a (e_\mu)^c] [\nabla_b (e_\nu)_c]. \quad (43)$$

Similarly,

$$(e_\mu)^c \nabla_b \nabla_a (e_\nu)_c = \nabla_b \omega_{a\mu\nu} - [\nabla_b (e_\mu)^c] [\nabla_a (e_\nu)_c]. \quad (44)$$

Subtracting (44) from (43) gives

$$\begin{aligned} (e_\mu)^c (\nabla_a \nabla_b - \nabla_b \nabla_a) (e_\nu)_c &= \nabla_a \omega_{b\mu\nu} - \nabla_b \omega_{a\mu\nu} \\ &\quad - [\nabla_a (e_\mu)^c] [\nabla_b (e_\nu)_c] + [\nabla_b (e_\mu)^c] [\nabla_a (e_\nu)_c]. \end{aligned} \quad (45)$$

Now expand the last two terms using the completeness relation

$$\delta^c_f = \eta^{\alpha\beta} (e_\alpha)^c (e_\beta)_f. \quad (46)$$

We obtain

$$\begin{aligned} [\nabla_a (e_\mu)^c] [\nabla_b (e_\nu)_c] &= [\nabla_a (e_\mu)^f] \delta^c_f [\nabla_b (e_\nu)_c] \\ &= \eta^{\alpha\beta} [\nabla_a (e_\mu)^f] (e_\alpha)_f (e_\beta)^c [\nabla_b (e_\nu)_c] \\ &= \eta^{\alpha\beta} \omega_{a\alpha\mu} \omega_{b\beta\nu}. \end{aligned} \quad (47)$$

Similarly,

$$[\nabla_b (e_\mu)^c] [\nabla_a (e_\nu)_c] = \eta^{\alpha\beta} \omega_{b\alpha\mu} \omega_{a\beta\nu}. \quad (48)$$

Substituting (47) and (48) into (45), we get

$$\begin{aligned} (e_\mu)^c (\nabla_a \nabla_b - \nabla_b \nabla_a) (e_\nu)_c &= \nabla_a \omega_{b\mu\nu} - \nabla_b \omega_{a\mu\nu} \\ &\quad - \eta^{\alpha\beta} \omega_{a\alpha\mu} \omega_{b\beta\nu} + \eta^{\alpha\beta} \omega_{b\alpha\mu} \omega_{a\beta\nu}. \end{aligned} \quad (49)$$

Finally, contracting with $(e_\rho)^a (e_\sigma)^b$ and using (42), we obtain

$$\begin{aligned} R_{\rho\sigma\mu\nu} &= (e_\rho)^a (e_\sigma)^b [\nabla_a \omega_{b\mu\nu} - \nabla_b \omega_{a\mu\nu} - \eta^{\alpha\beta} (\omega_{a\alpha\mu} \omega_{b\beta\nu} - \omega_{b\alpha\mu} \omega_{a\beta\nu})] \\ &= (e_\rho)^a \nabla_a [(e_\sigma)^b \omega_{b\mu\nu}] - (e_\sigma)^b \nabla_b [(e_\rho)^a \omega_{a\mu\nu}] \\ &\quad - (e_\rho)^a [\nabla_a (e_\sigma)^b] \omega_{b\mu\nu} + (e_\sigma)^b [\nabla_b (e_\rho)^a] \omega_{a\mu\nu} \\ &\quad - \eta^{\alpha\beta} [\omega_{\rho\beta\mu} \omega_{\sigma\alpha\nu} - \omega_{\sigma\beta\mu} \omega_{\rho\alpha\nu}] \\ &= (e_\rho)^a \nabla_a \omega_{\sigma\mu\nu} - (e_\sigma)^a \nabla_a \omega_{\rho\mu\nu} \\ &\quad - \eta^{\alpha\beta} \omega_{\alpha\beta\rho} \omega_{\sigma\mu\nu} + \eta^{\alpha\beta} \omega_{\alpha\beta\sigma} \omega_{\rho\mu\nu} \\ &\quad - \eta^{\alpha\beta} [\omega_{\rho\beta\mu} \omega_{\sigma\alpha\nu} - \omega_{\sigma\beta\mu} \omega_{\rho\alpha\nu}] \\ &= (e_\rho)^a \nabla_a \omega_{\sigma\mu\nu} - (e_\sigma)^a \nabla_a \omega_{\rho\mu\nu} \\ &\quad - \eta^{\alpha\beta} [\omega_{\rho\beta\mu} \omega_{\sigma\alpha\nu} - \omega_{\sigma\beta\mu} \omega_{\rho\alpha\nu} + \omega_{\alpha\beta\rho} \omega_{\sigma\mu\nu} - \omega_{\alpha\beta\sigma} \omega_{\rho\mu\nu}] \\ &= (e_\rho)^a \partial_a \omega_{\sigma\mu\nu} - (e_\sigma)^a \partial_a \omega_{\rho\mu\nu} \\ &\quad - \eta^{\alpha\beta} [\omega_{\rho\beta\mu} \omega_{\sigma\alpha\nu} - \omega_{\sigma\beta\mu} \omega_{\rho\alpha\nu} + \omega_{\alpha\beta\rho} \omega_{\sigma\mu\nu} - \omega_{\alpha\beta\sigma} \omega_{\rho\mu\nu}]. \end{aligned} \quad (50)$$

The components of the Ricci tensor may then be computed via the formula $R_{\rho\mu} = \eta^{\sigma\nu} R_{\rho\sigma\mu\nu}$.

4.2 Form Representation

Equations (50) and (24) can be expressed more elegantly in the notation of differential form, as stated in the following theorem.

Theorem 4.1. *In differential forms notation, the equation (24) becomes*

$$de_\sigma = e_\mu \wedge \omega_\sigma^\mu. \quad (51)$$

and equation (50) can be written as

$$R_\mu{}^\nu = d\omega_\mu{}^\nu + \omega_\mu{}^\alpha \wedge \omega_\alpha{}^\nu, \quad (52)$$

where $R_\mu{}^\nu = R_{ab\mu}{}^\nu = \eta^{\nu\rho} R_{ab\mu\rho}$, $\omega_\nu{}^\mu \equiv \eta^{\mu\sigma} \omega_{\nu\sigma}$.

Proof. (1) Equation (24) implies (51). In a coordinate basis, the exterior derivative of e_σ is

$$de_\sigma = 2\nabla_{[a}(e_\sigma)_{b]}. \quad (53)$$

Using (25) to substitute $\nabla_{[a}(e_\sigma)_{b]}$, we get

$$de_\sigma = 2\eta^{\mu\nu}(e_\mu)_{[a}\omega_{b]\sigma\nu} = \eta^{\mu\nu}e_\mu \wedge \omega_{\sigma\nu} = e_\mu \wedge \omega_\sigma^\mu, \quad (54)$$

which is (51).

(2) Equation (50) implies (52). Substituting (50) into $R_{ab\mu}{}^\nu = \eta^{\nu\rho}(e_\rho)^a(e_\sigma)^b R_{\rho\sigma\mu\nu}$, and using $\eta^{\mu\nu}(e_\mu)^a(e_\nu)_b = \delta^a_b$ gives

$$R_{ab\mu}{}^\nu = \partial_a\omega_{b\mu}{}^\nu - \partial_b\omega_{a\mu}{}^\nu + \omega_{a\mu}{}^\alpha\omega_{b\alpha}{}^\nu - \omega_{b\mu}{}^\alpha\omega_{a\alpha}{}^\nu. \quad (55)$$

Using the definitions of the exterior derivative:

$$d\omega_\mu{}^\nu = 2\partial_{[a}\omega_{b]\mu}{}^\nu = (\partial_a\omega_{b\mu}{}^\nu - \partial_b\omega_{a\mu}{}^\nu) \quad (G)$$

and the wedge product:

$$(\omega_\mu{}^\alpha \wedge \omega_\alpha{}^\nu)_{ab} = 2\omega_{[a|\mu}{}^\alpha\omega_{|b]\alpha}{}^\nu = \omega_{a\mu}{}^\alpha\omega_{b\alpha}{}^\nu - \omega_{b\mu}{}^\alpha\omega_{a\alpha}{}^\nu, \quad (H)$$

we can rewrite (55) as

$$R_\mu{}^\nu = d\omega_\mu{}^\nu + \omega_\mu{}^\alpha \wedge \omega_\alpha{}^\nu. \quad (56)$$

This is precisely (52). $\square$

References

- [1] Wald, R. M. *General Relativity*. Chicago: The University of Chicago Press, 1984.
- [2] Liang, C. B. and Zhou, B. *Introduction to Differential Geometry and General Relativity, Volume I*. 2nd ed. Beijing: Science Press, 2006.
- [3] DeepSeek. *DeepSeek [Large AI Model]*. 2024. URL: <https://www.deepseek.com> (Accessed: July 29, 2026).

A Appendix

A.1 Report on Use of AI Tools

The conceptual understanding and verification of specific ideas were developed through Q& A interactions with AI. AI tools were also used for English language polishing.

A.2 What Is New in This Work

This text is my reading notes on Wald's book [1], intended to supplement the specific computational details, while following the basic content of the original work.

Specifically, in Sec. 1, I supply the proof of Eq. (2); in Sec. 2, I provide a detailed proof of Theorem 2.1; in Sec. 3, following Liang's book [2], I prove Theorem 3.1 and supplement Lemma 3.2 together with its proof.

In Sec. 4.1, I explain the second equality in Eq. (42) and give a detailed derivation of Eq. (50). In Sec. 4.2, I provide a detailed proof of Theorem 4.1.

Finally, it should be emphasized that this note is merely a set of reading notes on Wald's book; the overall logic and ideas are inherited from Wald's work. My contribution consists only of rephrasing, reorganizing, and supplementing the specific computational and proof details. If there is anything of value in this note, the credit should first go to Wald's book. Should the copyright holder believe that this note infringes upon the copyright of Wald's book, please contact me and I will remove it promptly.